Marika Nishi Mechatronics – Diploma / Engineer

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EDUCATION

University of Pennsylvania, School of Engineering & Applied Science | Philadelphia, PA, United States *May 2027*
Candidate for Master of Science in Engineering, *Majors: Robotics, Computer Science* | *Concentration: Computer Vision* GPA: 3.78/4.0
University of Tokyo, Faculty of Engineering | Bunkyo, Tokyo, Japan *March 2023*
Bachelor of Engineering, *Major: Mechanical Engineering* GPA: 3.64/4.0

SKILLS & RELEVANT COURSEWORK

Software Skills: industrial communication protocols (CAN), SLAM, perception, programming, Python, PyTorch, C++, C, ROS, ROS2, SQL, sensor fusion, OpenCV, segmentation, ML-based planning, simulator (Gazebo, IsaacSim), autonomous driving
Hardware Skills: CAD, LiDAR, circuit design, RGB cameras, GPS, IMU, actuators, 3D printers, laser cutters, microcontroller
Relevant Coursework: LLM; Dynamics and Control; Mechatronics; F1tenth Autonomous Racing Cars; Automotive Engineering; Deep Neural Networks; Deep Learning; Computer Vision; Machine Perception; Machine Learning
Publication: Hori, K., Watanabe, K., Tanaka, T., **Nishi, M.**, & Ito, T. (2026). Visibility Evaluation Around Unsignalized Intersections with Occlusion From Arbitrary On-road Viewpoints Using Point Cloud Maps. *International Journal of Intelligent Transportation Systems Research*.
Watanabe, K., **Nishi, M.**, & Ito, T. (2025). Placement design of various roadside sensors for stochastic position prediction of traffic participants on community roads. *International Journal of Intelligent Transportation Systems Research*.

EMPLOYMENT EXPERIENCE

Vijay Kumar Lab | *Research Assistant,* Philadelphia, PA, United States *May 2026 – Present*

- Design and build VLA control system of drones with robotic arms to complete tasks like turning valves with IsaacSim

ISUZU | *Advanced Engineering - AI & Autonomous Driving Intern,* Plymouth, MI, United States *May 2026 – August 2026*

- Investigate autonomous truck on-road test failures by tracing issues across a large-scale Bazel-based perception, planning, and control software stack, identifying root causes through code analysis and debugging
- Use SQL and AWS to analyze autonomous driving on-road test data and quantitatively evaluate stack performance

Vijay Kumar Lab | *Research Intern,* Philadelphia, PA, United States *May 2025 – May 2026*

- Implemented driving simulators that visualized data of event camera on F1tenth car using Gazebo, CARLA, and ROS
- Collected datasets on driving simulator and trained vision transformer model that computed appropriate motion control from event camera images; improve data generation process efficiency by removing redundancy, cutting time by 70 %

BOSCH | *Software Intern*, Tsuzuki, Kanagawa, Japan *August 2023 – September 2023*

- Designed and built map to enhance autonomous driving/ADAS by visualizing key information, such as routes and vehicle state using MATLAB; helped company's engineers understand signals sent from cars visually
- Improved software that processed CAN signals from cars by fixing signal decoding; refined system output accuracy
- Collaborated with colleagues from 10 countries and different majors; made outcomes reflecting diverse opinions

DMG MORI | *Industrial Practices Intern, Additive Manufacturing R & D*, Iga, Mie, Japan *August 2021 – September 2021*

- Analyzed performance tests of additive manufacturing machines by operating the machines and inspecting products
- Investigated additive manufacturing methods by examining machine structures; proposed solutions to heat distortion

ROBOT AUTOMATION PROJECTS

Mobile Robots Competition | *Mechatronics, CAD, Mechanical Design, Microcontroller, Control, Webpage, C* *January 2026 – May 2026*

- Designed and fabricated a mobile robot with CAD, laser cutting, and 3D printing; wrote embedded C (Arduino) code for motor control, ToF wall-following, and wireless web interface using UDP and IP; won 2nd place in final competition

Research of Traffic Collision Prediction System | *Prototyping, Extended Kalman Filter , C++, ROS* *April 2022 – March 2023*

- Designed and built real-time collision prediction system among pedestrian, cyclist, and intelligent wheelchair
- Wrote C++ ROS codes that detected pedestrians and cyclists based on LiDAR data
- Conducted sensor fusion using Extended Kalman Filter; estimated pedestrians and cyclists' positions
- Operated RTK-GPS, low-accuracy GPS, LiDAR, and IMU to collect and integrate data into system development